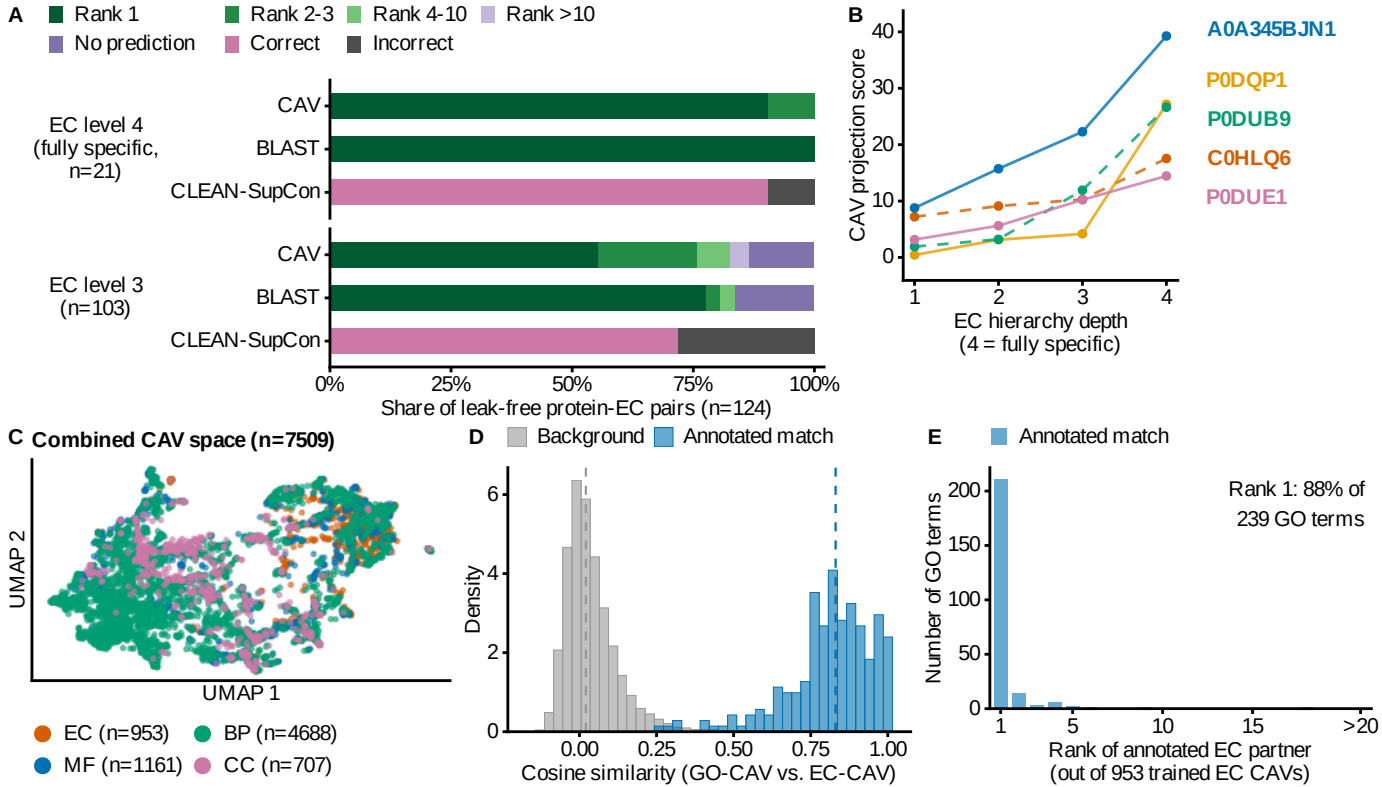

A
Rank 1
Rank 2-3
Rank 4-10
Rank >10
No prediction
Correct
Incorrect
EC level 4
(fully specific,
n=21)
CAV
BLAST
CLEAN-SupCon
EC level 3
(n=103)
CAV
BLAST
CLEAN-SupCon
0%
25%
50%
75%
100%
Share of leak-free protein-EC pairs (n=124)
B
CAV projection score
EC hierarchy depth
(4 = fully specific)
A0A345BJN1
P0DQP1
P0DUB9
C0HLQ6
P0DUE1
C
Combined CAV space (n=7509)
UMAP 2
UMAP 1
EC (n=953)
BP (n=4688)
MF (n=1161)
CC (n=707)
D
Background
Annotated match
Density
Cosine similarity (GO-CAV vs. EC-CAV)
E
Annotated match
Number of GO terms
Rank 1: 88% of
239 GO terms
Rank of annotated EC partner
(out of 953 trained EC CAVs)